```
[NbConvertApp] Converting notebook Doc.ipynb to webpdf
[NbConvertApp] WARNING | Alternative text is missing on 4 image(s).
[NbConvertApp] Building PDF
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 3128817 bytes to Doc.pdf
```

Testing LightGBM & XGBoost C APIs on Edge Devices

Engine Fault Detection (Classification) & Windmill Power Prediction (Regression)

Target Devices Specifications

Raspberry Pi Zero 2 W

- CPU: Broadcom BCM2710A1 (Quad-core 64-bit ARM Cortex-A53 @ 1GHz)
- RAM: 512MB LPDDR2
- Storage: MicroSD card (OS-dependent) Alpine Linux OS

Luckfox Lyra Plus

- CPU: Rockchip RV1103 (Dual-core ARM Cortex-A7 @ 1.2GHz)
- RAM: 256MB DDR3
- Storage: SPI Flash (OS-dependent) RockchipBuildRoot OS

Compilation

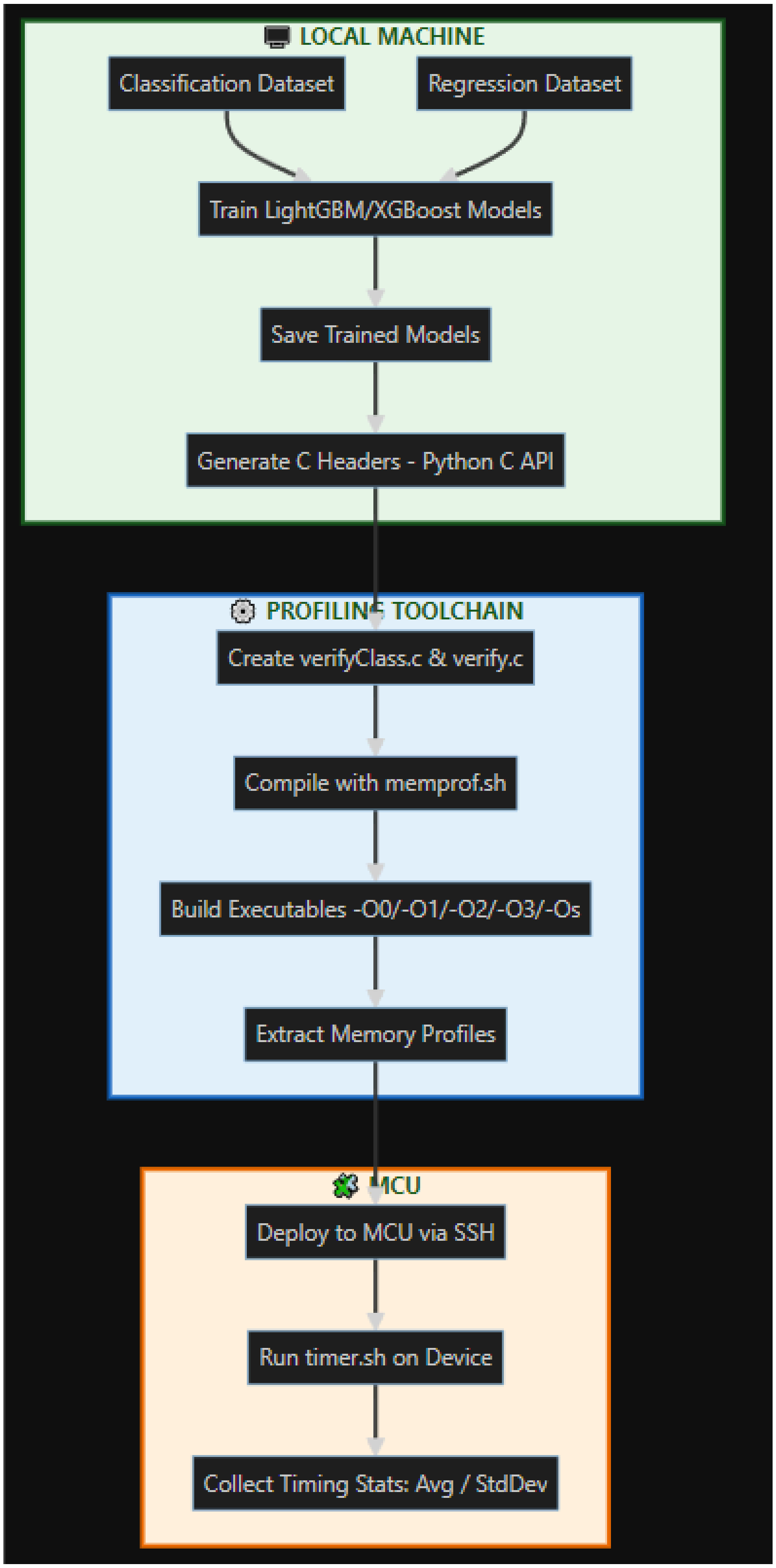
Luckfox Lyra Plus is done in arm-rockchip830-linux-uclibcgnueabihf-gcc **Raspberry Pi Zero 2 W** is done in arm-linux-gnueabihf-gcc

Test Objectives

1. Validate C API inference performance for classification/regression
2. Measure memory footprint on resource-constrained devices
3. Verify cross-compilation compatibility for ARM architectures

---

WORKFLOW



🧪 Example

💻 Compilation on Local Machine

The classification model verification code was compiled for ARM architecture using arm-linux-gnueabihf-gcc with static linking and no optimization (-O0). The resulting binary (verify) was analyzed for memory footprint using arm-linux-gnueabihf-size. Key sections include:

```
martini@martini-Virtual-Machine:~/shared-drives/G:/My Drive/WORK/TREES/2Board$ arm-linux-gnueabihf-gcc -static -O0  verifyClass
.c -o verify -lm
martini@martini-Virtual-Machine:~/shared-drives/G:/My Drive/WORK/TREES/2Board$ arm-linux-gnueabihf-size -A verify
verify  :
section                  size     addr
.note.gnu.build-id         36    65812
.note.ABI-tag              32    65848
.rel.dyn                   16    65880
.init                      12    65896
.iplt                      32    65908
.text                  258992    65984
__libc_freeres_fn        1964   324976
.fini                       8   326940
.rodata                410576   326952
.ARM.extab                504   737528
.ARM.exidx               1440   738032
.eh_frame                   4   739472
.tdata                     16   807064
.tbss                      36   807080
.init_array                 4   807080
.fini_array                 4   807084
.data.rel.ro             8016   807088
.got                      604   815104
.data                    7380   815712
__libc_subfreeres          36   823092
__libc_IO_vtables         840   823128
__libc_atexit               4   823968
.bss                    11000   823976
__libc_freeres_ptrs        16   834976
.comment                   43        0
.ARM.attributes            53        0
Total                  701668
```

```
martini@martini-Virtual-Machine:~/shared-drives/G:/My Drive/WORK/TREES/2Board$ arm-linux-gnueabihf-size -A verify | grep -E '\.
rodata|\.data'
.rodata                410576   326952
.data.rel.ro             8016   807088
.data                    7380   815712
```

🔁 Transfer to Microcontroller (MCU)

Copy and send via ssh the compiled model to your microcontroller using your preferred tool:

```
martini@martini-Virtual-Machine:~/shared-drives/G:/My Drive/WORK/TREES/2Board$ scp verify stavros@172.20.10.13:/home/stavros/
stavros@172.20.10.13's password:
verify                                                                          100%  776KB   2.8MB/s   00:00
```

📈 Inference Results on Target

Check the prediction output after running inference on the embedded device:

```
alpinepi:~$ ./verify
Sample      Class 0   Class 1   Class 2   Class 3
-------------------------------------------------
0           0.999967  0.000000  0.000018  0.000015
1           0.000104  0.000082  0.462228  0.537586
2           0.999962  0.000000  0.000018  0.000020
3           0.000000  0.999909  0.000058  0.000033
4           0.999933  0.000004  0.000048  0.000015
5           0.000113  0.000088  0.414069  0.585730
6           0.999986  0.000004  0.000006  0.000004
7           0.000093  0.000071  0.473825  0.526011
8           0.000093  0.000073  0.491760  0.508074
9           0.000102  0.000082  0.505711  0.494104
10          0.000093  0.000071  0.485209  0.514627
11          0.000000  0.999906  0.000048  0.000045
12          0.000097  0.000079  0.500822  0.499002
13          0.000106  0.000083  0.516088  0.483722
14          0.000106  0.000083  0.554671  0.445141
15          0.999966  0.000000  0.000028  0.000007
16          0.999968  0.000000  0.000020  0.000012
17          0.000000  0.999828  0.000096  0.000076
18          0.000100  0.000077  0.509836  0.489987
19          0.999956  0.000000  0.000026  0.000018
20          0.000000  0.999962  0.000024  0.000014
21          0.000095  0.000074  0.474278  0.525553
22          0.000099  0.000074  0.499882  0.499946
23          0.000000  0.999962  0.000021  0.000017
24          0.999968  0.000000  0.000011  0.000021
25          0.999991  0.000000  0.000004  0.000005
26          0.000098  0.000081  0.486907  0.512914
27          0.000108  0.000082  0.489052  0.510758
28          0.000091  0.000069  0.536522  0.463318
29          0.000083  0.000065  0.454639  0.545213
30          0.000000  0.999974  0.000009  0.000016
31          0.000100  0.000075  0.489471  0.510353
32          0.000105  0.000083  0.555422  0.444389
33          0.999986  0.000004  0.000006  0.000004
34          0.000102  0.000077  0.538132  0.461689
35          0.999966  0.000003  0.000014  0.000017
36          0.000000  0.999983  0.000011  0.000006
37          0.000099  0.000076  0.526205  0.473620
38          0.999946  0.000000  0.000030  0.000024
39          0.000000  0.999887  0.000057  0.000056
Elapsed time (40 Predictions): 9514846 nsec
Meaning 237871 nsec/prediction
```

memprof.sh Script

We made a bash script to check how much memory our machine learning code uses. Here's why it's useful:

What the Script Does:

1. Compiles the code 5 times with different optimizations (-O0 ,-O1 , -O2, -O3 and -Os)
2. Measures:
   - Flash memory needed (code + constants)
   - RAM needed (variables)
3. Shows results in an easy-to-read table

(CLASSIFICATION)

RASPBERRY PI LightGBM

🛠️ Compilation Output

```
=== Building with -O0 ===

Memory Report for build_0:
Build 0 (-O0):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 311 KB
  Total Flash: 313 KB

RAM Usage:
  .data   (Vars): 3 KB
  Total RAM:  3 KB
--------------------------------

=== Building with -O1 ===

Memory Report for build_1:
Build 1 (-O1):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 311 KB
  Total Flash: 312 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
--------------------------------

=== Building with -O2 ===

Memory Report for build_2:
Build 2 (-O2):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 311 KB
  Total Flash: 312 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
```

```
=== Building with -O3 ===

Memory Report for build_3:
Build 3 (-O3):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 311 KB
  Total Flash: 312 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
--------------------------------

=== Building with -Os ===

Memory Report for build_4:
Build 4 (-Os):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 310 KB
  Total Flash: 311 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
--------------------------------

Done! Created executables:
-rwx------ 1 martini martini 338K Iouv  2 01:19 build_0
-rwx------ 1 martini martini 336K Iouv  2 01:20 build_1
-rwx------ 1 martini martini 336K Iouv  2 01:20 build_2
-rwx------ 1 martini martini 336K Iouv  2 01:20 build_3
```

🔍 Info Matching (C-API Verification)

Confirming that the C API correctly reflects the expected structure and behavior:

```
Tree Stats:
  Total Trees         : 400
  Max Depth           : 19
  Min Depth           : 8
  Avg Depth           : 11.30
  Depth Std Dev       : 2.03
  Total Nodes         : 24400
  Max Nodes Per Tree  : 61

Estimated Memory Usage: 311.33 KB
```

RASPBERRY PI XGBoost

```
Memory Report for build_0:
Build 0 (-O0):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 333 KB
  Total Flash: 335 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
--------------------------------

=== Building with -O1 ===
Linkage: Dynamic

Memory Report for build_1:
Build 1 (-O1):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 333 KB
  Total Flash: 334 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
--------------------------------

=== Building with -O2 ===
Linkage: Dynamic

Memory Report for build_2:
Build 2 (-O2):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 333 KB
  Total Flash: 334 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
```

```
=== Building with -O3 ===
Linkage: Dynamic

Memory Report for build_3:
Build 3 (-O3):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 333 KB
  Total Flash: 334 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
--------------------------------

=== Building with -Os ===
Linkage: Dynamic

Memory Report for build_4:
Build 4 (-Os):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 333 KB
  Total Flash: 334 KB

RAM Usage:
  .data   (Vars): 2 KB
  Total RAM:  2 KB
--------------------------------

Done! Created executables:
360K build_0
360K build_1
360K build_2
360K build_3
360K build_4
```

```
    Tree Stats:
      Total Trees        : 400
      Max Depth          : 7
      Min Depth          : 2
      Avg Depth          : 6.14
      Depth Std Dev      : 1.46
      Total Nodes        : 21308
      Max Nodes Per Tree : 117

    Estimated Memory Usage: 499.41 KB
```

```
alpinepi:~$ sh timer.sh
+------+---------------+----------------+
| Opt  | Avg Time (ns) | Std Dev (ns)   |
+------+---------------+----------------+
| O0   | 216550        | 27106          |
| O1   | 92876         | 127529         |
| O2   | 95258         | 125909         |
| O3   | 96613         | 125372         |
| Os   | 124605        | 112681         |
+------+---------------+----------------+
```

REGREESION

LIGHTGBM RASPBERRY PI

```
Memory Report for build_0:
Build 0 (-O0):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 120 KB
  Total Flash: 122 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------

=== Building with -O1 ===
Linkage: Dynamic

Memory Report for build_1:
Build 1 (-O1):
--------------------------------
Flash Usage:
  .text   (Code): 0 KB
  .rodata (Const): 120 KB
  Total Flash: 121 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------

=== Building with -O2 ===
Linkage: Dynamic

Memory Report for build_2:
Build 2 (-O2):
--------------------------------
Flash Usage:
  .text   (Code): 0 KB
  .rodata (Const): 120 KB
  Total Flash: 121 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
```

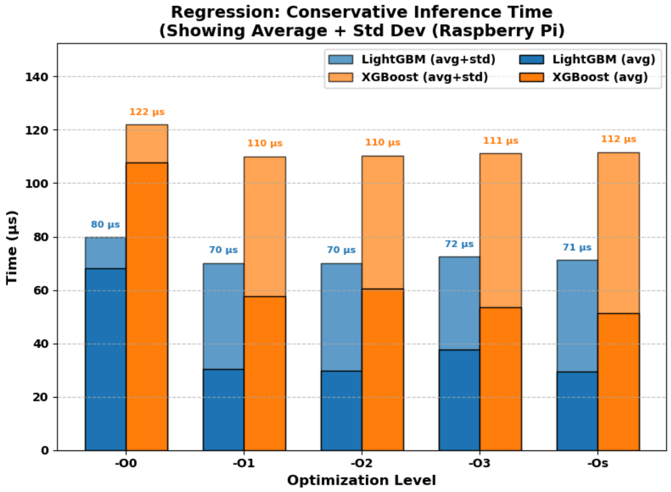
```
=== Building with -O3 ===
Linkage: Dynamic

Memory Report for build_3:
Build 3 (-O3):
--------------------------------
Flash Usage:
  .text   (Code): 0 KB
  .rodata (Const): 120 KB
  Total Flash: 121 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------

=== Building with -Os ===
Linkage: Dynamic

Memory Report for build_4:
Build 4 (-Os):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 120 KB
  Total Flash: 121 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------

Done! Created executables:
137K build_0
136K build_1
136K build_2
136K build_3
137K build_4
```

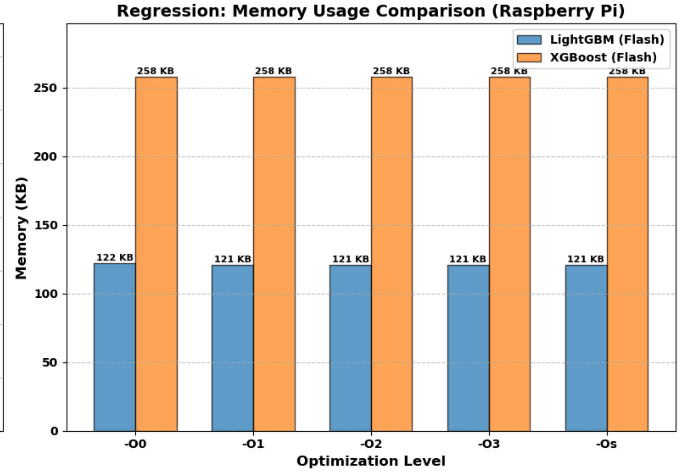
```
alpinepi:~$ sh timer.sh
+------+---------------+---------------+
| Opt  | Avg Time (ns) | Std Dev (ns)  |
+------+---------------+---------------+
| O0   | 68149         | 11655         |
| O1   | 30509         | 39561         |
| O2   | 29854         | 40240         |
| O3   | 37642         | 34824         |
| Os   | 29276         | 41912         |
+------+---------------+---------------+
```

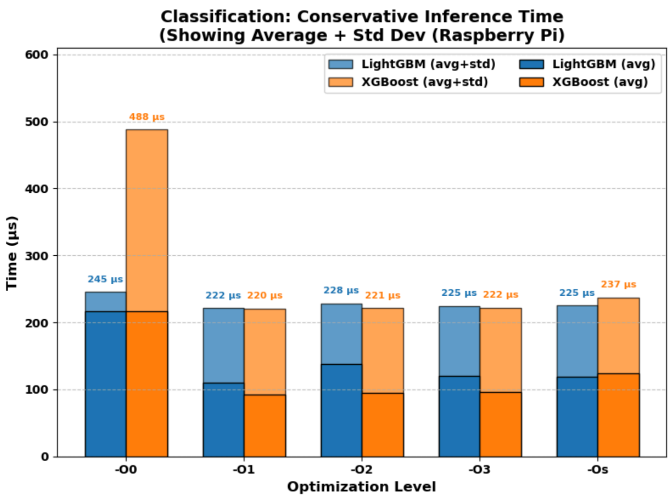
XGBOOST RASPVERY PI

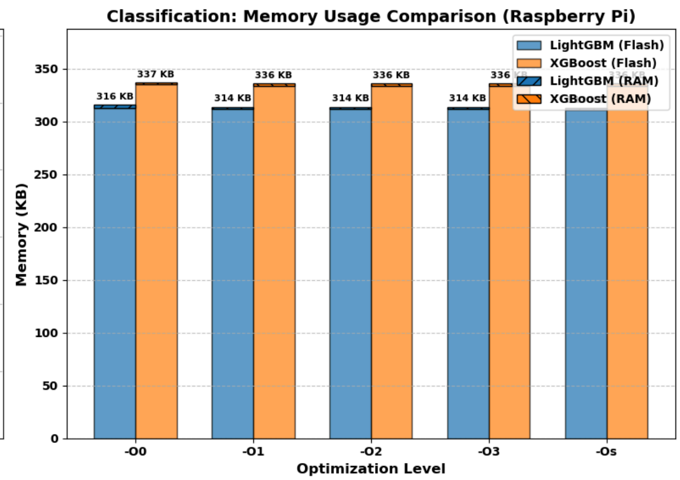
```
Build 0 (-O0):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 257 KB
  Total Flash: 258 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------

=== Building with -O1 ===
Linkage: Dynamic

Memory Report for build_1:
Build 1 (-O1):
--------------------------------
Flash Usage:
  .text   (Code): 0 KB
  .rodata (Const): 257 KB
  Total Flash: 258 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------

=== Building with -O2 ===
Linkage: Dynamic

Memory Report for build_2:
Build 2 (-O2):
--------------------------------
Flash Usage:
  .text   (Code): 0 KB
  .rodata (Const): 257 KB
  Total Flash: 258 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------
```

```
=== Building with -O3 ===
Linkage: Dynamic

Memory Report for build_3:
Build 3 (-O3):
--------------------------------
Flash Usage:
  .text   (Code): 0 KB
  .rodata (Const): 257 KB
  Total Flash: 258 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------

=== Building with -Os ===
Linkage: Dynamic

Memory Report for build_4:
Build 4 (-Os):
--------------------------------
Flash Usage:
  .text   (Code): 1 KB
  .rodata (Const): 257 KB
  Total Flash: 258 KB

RAM Usage:
  .data   (Vars): 0 KB
  Total RAM:  0 KB
--------------------------------

Done! Created executables:
273K build_0
272K build_1
272K build_2
272K build_3
273K build_4

alpinepi:~$ sh timer.sh
+------+----------------+----------------+
| Opt  | Avg Time (ns)  | Std Dev (ns)   |
+------+----------------+----------------+
| O0   | 107732         | 14185          |
| O1   | 57569          | 52550          |
| O2   | 60671          | 49869          |
| O3   | 53529          | 57633          |
| Os   | 51324          | 60379          |
+------+----------------+----------------+
```

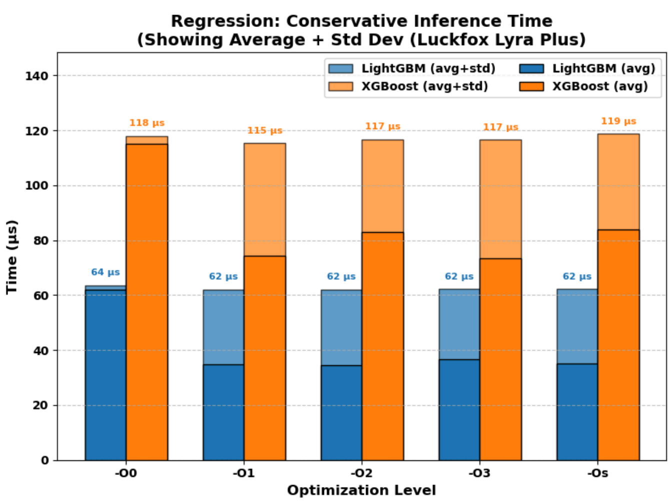


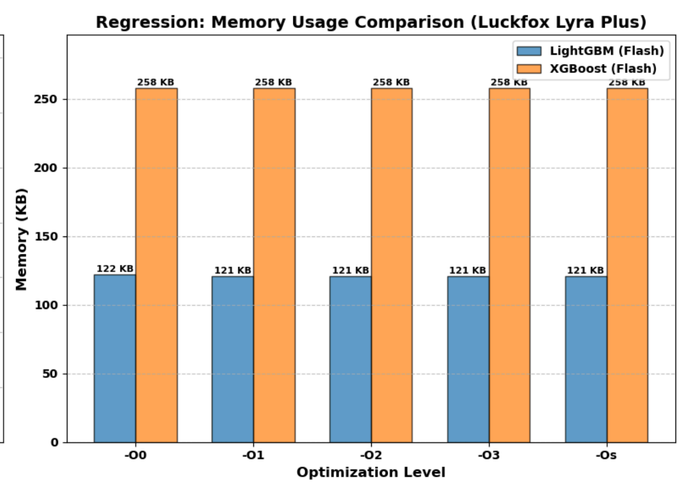


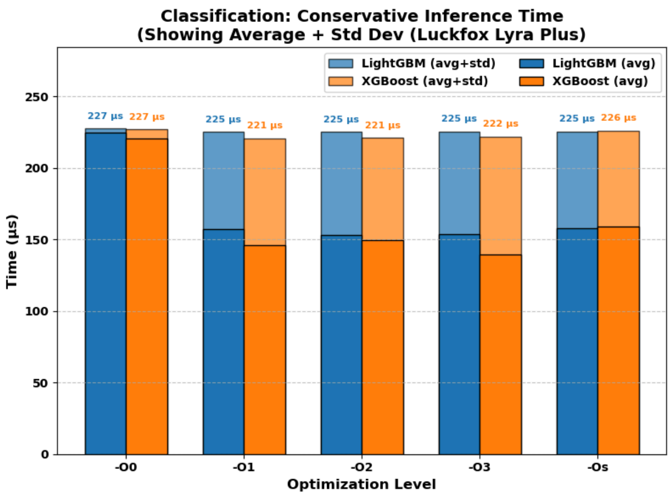


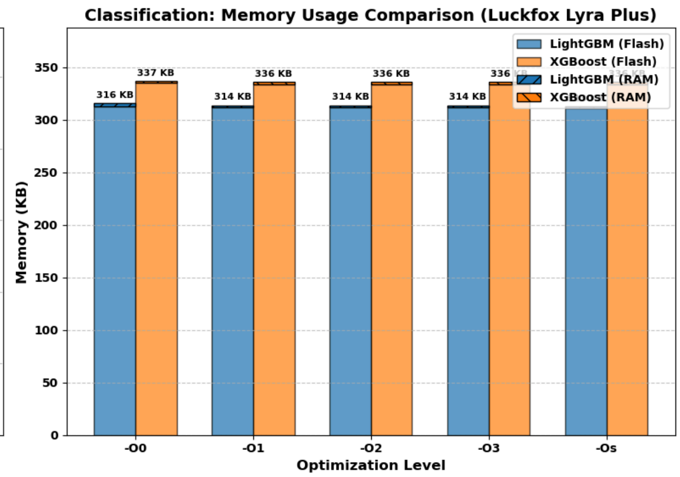


Same procees is also repeded for the different board and the results gotten are:






Classification: Conservative Inference Time
(Showing Average + Std Dev (Luckfox Lyra Plus)
LightGBM (avg+std)
XGBoost (avg+std)
LightGBM (avg)
XGBoost (avg)
227 µs
227 µs
225 µs
221 µs
225 µs
221 µs
225 µs
222 µs
225 µs
226 µs
Time (µs)
Optimization Level
-O0
-O1
-O2
-O3
-Os


Classification: Memory Usage Comparison (Luckfox Lyra Plus)
LightGBM (Flash)
XGBoost (Flash)
LightGBM (RAM)
XGBoost (RAM)
316 KB
337 KB
314 KB
336 KB
314 KB
336 KB
314 KB
Memory (KB)
Optimization Level
-O0
-O1
-O2
-O3
-Os